Evaluation 1 — Tiny Worlds: An Ant-Scale XR Insect Discovery Game

1. Objective and Validation Metrics

Prototype 1 tested whether first-time users could move from insect selection into ladybird exploration, discover interactive anatomical features with minimal assistance, and understand the prototype's interaction language through hover enlargement, blue highlights, animation and contextual learning cards. A secondary aim was to investigate whether the spatial reveal of the elytra and hidden flight wings communicated meaningful biological information.

Success was evaluated using task completion, assistance required, wrong-control attempts, feature discovery, knowledge recall and 1–5 ratings for interaction clarity, blue-highlight clarity, information-card clarity, next-step clarity and overall ease of use.

2. Results

Measure	Result	What happened
Selection success	5/5	All participants entered the ladybird selection flow.
Feature discovery	Antenna 5/5; Elytra 4/5; Wings 4/5	Core anatomy was usually found, but not universally.
Wing sequence completed	3/5	Two participants did not complete the full wing interaction sequence.
Assistance	3/5 small hint; 2/5 multiple hints	Every participant received at least one hint in the recorded session.
Wrong control attempts	12 total (2.4 per participant)	Control mapping created avoidable interaction friction.
Knowledge recall	Antenna 1/5; Elytra 1/5; Wings 5/5 clearly correct	Recall was weak for card-dependent knowledge.

Knowledge scores use clearly correct responses from the observation forms. The wing result is interpreted cautiously because "wings help insects fly" may reflect prior knowledge.

Participant Experience Ratings

Statement (1–5)	Mean
I understood how to interact with the prototype	4.2
Blue highlight clearly showed what I was pointing at	3.8
Information cards were easy to understand	4.6
It was clear what I should do next	3.2
Overall, the prototype was easy to use	3.6

3. Synthesis of Results

- Theme 1 — Information was visible but not attended to. The cards received the strongest rating for clarity (4.6/5), yet antenna and elytra recall were both only 1/5. P01, P04 and P05 explicitly reported not wanting to read, not noticing, or seeing the card without reading it.
- Theme 2 — Users often lacked a clear next step. Next-step clarity was the lowest rating (3.2/5). P01 described a missing "mission", P02 asked what to do after an interaction, and P04 hesitated at the start of ladybird exploration.
- Theme 3 — The basic interaction language was promising but not fully self-explanatory. Selection succeeded for all participants and most anatomy was discovered, but blue-highlight clarity averaged 3.8/5, all participants received hints, and 12 wrong-control attempts were recorded.

4. Analysis/Insights

Insight A — The educational problem is attention, not text readability

The strongest contradiction in the data is that participants rated the information cards as easy to understand (4.6/5) while failing to retain the card-dependent concepts. Only one participant clearly recalled the function of the antennae and only one recalled the elytra. This indicates that simplifying the wording alone is unlikely to solve the problem. During active exploration, users prioritized movement, animation and target finding over voluntary reading. The text card therefore competed with the interaction rather than becoming part of it.

Design implication: Move from text-only delivery to short multimodal micro-learning. On the first discovery of a body part, play a brief spoken explanation synchronized with the animation while retaining a concise caption and optional replay. Treat audio as a hypothesis to test, not something assumed to improve learning automatically.

Insight B — The experience needs progressive guidance without removing discovery

Participants could usually operate the prototype once they understood the controls, but several were uncertain about what should happen next. The low next-step rating (3.2/5), repeated hints, and comments about missing missions suggest that the experience currently relies too heavily on users inferring the sequence themselves. However, fully instructing users to click specific anatomy would undermine the exploratory character of the concept.

Design implication: Introduce lightweight progressive guidance: a short goal at the start of exploration (for example, "Discover how the ladybird senses and flies"), consistent control onboarding, and contextual hints only after inactivity. A subtle discovery count can indicate progress without naming the next body part.

Insight C — Spatial and visual affordances are useful, but consistency and visibility matter

The interaction system was broadly discoverable: all participants found the antennae and four of five found both elytra and wings. This supports keeping spatial interaction, animation, and blue hover feedback. However, the blue cue was not equally clear for everyone (3.8/5), and P02 specifically reported that selected content was unclear and that cards should respond to viewing angle. P04 also reported that the initial ladybird view felt too close.

Design implication: Retain the blue interaction language but apply it consistently to both 3D anatomy and UI buttons, orient learning content toward the user, and adjust the initial viewing distance so interactive parts are readable and targetable.

5. Evaluation of Aims

Testing aim	Evaluation	Evidence / reasoning
First-time users can navigate the flow with minimal assistance	Partially validated	5/5 selected successfully, but all recorded sessions required at least a small hint; next-step clarity averaged 3.2/5.
Visual and behavioural cues make anatomy discoverable	Mostly validated	Antennae 5/5, elytra 4/5 and wings 4/5 discovered; blue-highlight clarity averaged 3.8/5 rather than uniformly high.
The prototype communicates meaningful biological information	Not validated for antennae/elytra	Only 1/5 clearly recalled each card-dependent function. High card clarity did not translate into attention or recall.
Spatial interaction explains the elytra–wing relationship	Uncertain	The questionnaire asked separate function questions rather than directly testing the spatial relationship, so this aim was not adequately measured.

Spatial interaction explains the elytra–wing relationship is uncertain; this gap means Prototype 2 will add dedicated questions to measure users' spatial understanding of the elytra-wing physical relationship.

6. Concept Iteration

The next iteration keeps the ant-scale spatial exploration and staged anatomy reveal, but changes the experience from text-led discovery to guided multimodal discovery. The revised concept is designed to preserve curiosity while reducing dependence on voluntary reading and reducing uncertainty about progression.

Finding	Iteration	Expected effect / next test
Cards are clear but frequently ignored	Add 3–5 second narration on first discovery of antennae, elytra and flight wings; keep a short caption and a replay option.	Test whether narration and caption improves recall compared with text-only delivery without becoming intrusive.
Users are unsure what to do next	Add a simulator onboarding card, a high-level exploration goal, and delayed contextual hints after inactivity.	Measure assistance level, wrong-control attempts and next-step clarity again.
Blue cue works for most, not all	Standardise the same blue hover feedback across anatomy and UI buttons; preserve delayed card hiding to prevent flicker.	Check whether feature discovery and blue-cue rating become more consistent.
Card/viewing position can be unclear	Make learning cards face the user/camera and keep them outside the main anatomy sightline.	Observe whether participants notice information without losing focus on the insect.
Initial ladybird view felt too close for one participant	Increase initial viewing distance and preserve free movement around the model.	Observe targeting comfort and whether users still perceive the intended ant-scale effect.

Updated Concept Description

Tiny Worlds remains an ant-scale XR insect discovery experience, but the learning layer becomes multimodal and progressively guided. Users select an insect, enter a close-up habitat, and discover anatomical structures through spatial hover or select interactions. First-time discovery triggers a coordinated visual response, animation, short spoken explanation, and concise caption. Blue feedback remains the consistent signal for interactability, while lightweight goals and contextual prompts help users continue exploring without revealing every answer in advance.

7. Reflection and Next Steps

The mixed-method test was effective because it exposed a gap that a usability rating alone would have hidden: participants rated the information cards highly for clarity, yet frequently ignored them and showed weak recall. Combining observation, knowledge questions, ratings, and comments therefore produced a more useful evaluation than any single measure.

The methodology also had important limitations. The knowledge check used simple correct/incorrect responses and did not directly test the intended spatial relationship between elytra and wings. The wing question was also a weak educational measure because participants may already know that wings are used for flying. In the next test, I will add a direct relational question (for example, “Where are the flight wings when the ladybird is not flying, and what protects them?”), record confidence, and include a short delayed-recall check. A pre-test or comparison condition would also help separate prior knowledge from learning caused by the prototype.

Finally, this Week 5 session primarily evaluated first-use usability with peer participants rather than the intended child audience. The next prototype should therefore first verify the revised audio/guidance system with another small usability test, then evaluate it with appropriate target users before making claims about educational effectiveness. The key question for Prototype 2 is no longer simply whether users can operate the experience, but whether multimodal guidance improves attention and biological understanding while preserving exploratory agency. Key metrics will include knowledge recall scores, next-step clarity rating, amount of assistance required, and participants’ ability to describe spatial relationships between elytra and wings.

Appendix:

GitHub link: https://github.com/ChunyiZhang27/XR-Tracy_Zhang

Questionnaire & Observation Sheets:

DECO7230 Prototype 1 Usability Test

Prototype Usability Testing Questionnaire & Observation Sheet	Participant ID P01

The participant completes Sections B and C after the task. The researcher and participant complete Sections A and D.

A. Task Performance — Researcher observation

Measure	Record
Selection successful?	☐ Yes
Assistance required	☐ 2 Multiple hints
Antenna discovered?	☐ Yes
Elytra discovered?	☐ Yes
Wing discovered?	☐ Yes
Wing sequence completed?	☐ Yes

B. Knowledge Check — Researcher records the participant’s answer, then marks correctness

Question	Participant response	Correct?
What is the main function of an insect’s antennae?	Didn’ t notice that	☐ No
What is the function of the elytra?	Didn’ t notice that	☐ No
What is the function of the wings?	Help the insect fly	☐ Yes

C. Participant Experience Questionnaire — Circle one number for each statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood how to interact with the prototype.	○	○	○	4	○
The blue highlight clearly showed what I was pointing at.	○	○	○	○	5
The information cards were easy to understand.	○	○	○	○	5
It was clear what I should do next.	○	○	3	○	○
Overall, the prototype was easy to use.	○	○	○	4	○

D. Observation & Quote — Researcher notes hesitation, confusion, behaviours and memorable comments

Wrong control attempts how many times?	3
Where did the participant hesitate?	Sometimes don’t know what to do next
What caused confusion?	Info is missing, the mission is missing
Interesting behaviour / strategy	The movement of the prototype
Participant quote / other comments	“ More suitable for kids, don’t want to read the learning card ”

Prototype Usability Testing Questionnaire & Observation Sheet	Participant ID P02

The participant completes Sections B and C after the task. Researcher and participant complete Sections A and D.

A. Task Performance — Researcher observation

Measure	Record
Selection successful?	☐ Yes
Assistance required	☐ 1 Small hint
Antenna discovered?	☐ Yes
Elytra discovered?	☐ Yes
Wing discovered?	☐ Yes
Wing sequence completed?	☐ Yes

B. Knowledge Check — Researcher records the participant's answer, then marks correctness

Question	Participant response	Correct?
What is the main function of an insect's antennae?	Look, acquire relevant knowledge	☐ No
What is the function of the elytra?	Protection, from attack	☐ Yes
What is the function of the wings?	Fly	☐ Yes

C. Participant Experience Questionnaire — Circle one number for each statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood how to interact with the prototype.	○	○	○	○	5
The blue highlight clearly showed what I was pointing at.	○	○	3	○	○
The information cards were easy to understand.	○	○	○	4	○
It was clear what I should do next.	○	2	○	○	○
Overall, the prototype was easy to use.	○	○	○	4	○

D. Observation & Quote — Researcher notes hesitation, confusion, behaviours and memorable comments

Wrong control attempts how many times?	0
Where did the participant hesitate?	Control hints， What's the next step after operating?
What caused confusion?	Not many hints
Interesting behaviour / strategy	/
Participant quote / other comments	" The selected content is not very clear, even with hints, and the content cards should change according to the viewing angle. "

Prototype Usability Testing Questionnaire & Observation Sheet

Participant ID
P03

The participant completes Sections B and C after the task. Researcher and participant complete Sections A and D.

A. Task Performance — Researcher observation

Measure	Record
Selection successful?	☐ Yes
Assistance required	☐ 1 Small hint
Antenna discovered?	☐ Yes
Elytra discovered?	☐ Yes
Wing discovered?	☐ Yes
Wing sequence completed?	☐ Yes

B. Knowledge Check — Researcher records the participant's answer, then marks correctness

Question	Participant response	Correct?
What is the main function of an insect's antennae?	Shaking	☐ No
What is the function of the elytra?	Don't know	☐ No
What is the function of the wings?	Fly	☐ Yes

C. Participant Experience Questionnaire — Circle one number for each statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood how to interact with the prototype.	○	○	○	4	○
The blue highlight clearly showed what I was pointing at.	○	○	○	4	○
The information cards were easy to understand.	○	○	○	○	5
It was clear what I should do next.	○	○	○	4	○
Overall, the prototype was easy to use.	○	○	3	○	○

D. Observation & Quote — Researcher notes hesitation, confusion, behaviours and memorable comments

Wrong control attempts how many times?	5
Where did the participant hesitate?	When it comes to the interaction part, he doesn't know which button to confirm
What caused confusion?	Confused about use which letter to control which movement or how to confirms.
Interesting behaviour / strategy	The user would love to move around the insect, even enter the body.
Participant quote / other comments	" the prototype 1 is roughly to use "

Prototype Usability Testing Questionnaire & Observation Sheet	Participant ID P04

The participant completes Sections B and C after the task. Researcher and participant complete Sections A and D.

A. Task Performance — Researcher observation

Measure	Record
Selection successful?	☐ Yes
Assistance required	☐ 2 Multiple hints
Antenna discovered?	☐ Yes
Elytra discovered?	☐ No
Wing discovered?	☐ No
Wing sequence completed?	☐ No

B. Knowledge Check — Researcher records the participant's answer, then marks correctness

Question	Participant response	Correct?
What is the main function of an insect's antennae?	Wasn't paying attention	☐ Yes ☐ No
What is the function of the elytra?	Wasn't paying attention: read text but forgot	☐ Yes ☐ No
What is the function of the wings?	To fly	☐ Yes ☐ No

C. Participant Experience Questionnaire — Circle one number for each statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood how to interact with the prototype.	○	○	3	○	○
The blue highlight clearly showed what I was pointing at.	○	2	○	○	○
The information cards were easy to understand.	○	○	○	4	○
It was clear what I should do next.	○	○	3	○	○
Overall, the prototype was easy to use.	○	2	○	○	○

D. Observation & Quote — Researcher notes hesitation, confusion, behaviours and memorable comments

Wrong control attempts how many times?	4
Where did the participant hesitate?	The start of the lady bird discovery part.
What caused confusion?	It was too close when enter the ladybird page.
Interesting behaviour / strategy	Willing to use wads to move freely.
Participant quote / other comments	" I didn't notice the knowledge on the info card "

Prototype Usability Testing Questionnaire & Observation Sheet	Participant ID P05

The participant completes Sections B and C after the task. Researcher and participant complete Sections A and D.

A. Task Performance — Researcher observation

Measure	Record
Selection successful?	☐ Yes
Assistance required	☐ 1 Small hint
Antenna discovered?	☐ Yes
Elytra discovered?	☐ Yes
Wing discovered?	☐ Yes
Wing sequence completed?	☐ No

B. Knowledge Check — Researcher records the participant's answer, then marks correctness

Question	Participant response	Correct?
What is the main function of an insect's antennae?	Smell	☐ Yes
What is the function of the elytra?	Didn't notice that.	☐ No
What is the function of the wings?	Fly	☐ Yes

C. Participant Experience Questionnaire — Circle one number for each statement

1 = Strongly disagree 2 = Disagree 3 = Neutral 4 = Agree 5 = Strongly agree

Statement	1	2	3	4	5
I understood how to interact with the prototype.	○	○	○	○	5
The blue highlight clearly showed what I was pointing at.	○	○	○	○	5
The information cards were easy to understand.	○	○	○	○	5
It was clear what I should do next.	○	○	○	4	○
Overall, the prototype was easy to use.	○	○	○	○	5

D. Observation & Quote — Researcher notes hesitation, confusion, behaviors and memorable comments

Wrong control attempts how many times?	0
Where did the participant hesitate?	No hesitate.
What caused confusion?	No confusion appeared.
Interesting behaviour / strategy	Kind of familiar with the system.
Participant quote / other comments	" I see the info card but I didn't read it. "